

446 Lab 5

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Lab 5 Deliverables

1. Raw waveform is hard for the model to interpret directly because it is just amplitude over time. Spectrogram/MFCC features show **which frequencies are present and when**, which makes speech patterns easier to recognize. MFCCs also compress the audio into features that better match how humans perceive sound.
2. The representative dataset gives TensorFlow Lite sample inputs so it can learn the expected range of activation values. This lets the converter choose proper INT8 scale and zero-point values. Without it, the quantized model may have poor accuracy because the integer ranges may not match real audio features.
3. `audio_processor.get_data()`— loads and prepares audio examples for training, validation, or testing. In `input_data.py`, it usually handles selecting the correct audio files, applying preprocessing, creating spectrogram/MFCC-style input features, and returning the feature data with the correct labels for the model.
4. `g_model`— is the actual TensorFlow Lite model stored as a C/C++ byte array. `g_model_len` is the length of that array in bytes. They must be copied correctly into the Arduino source file because the microcontroller loads the model directly from that array. If the bytes or length are wrong, the model may fail to initialize or give incorrect predictions.



The screenshot shows the 'Serial Monitor' window of an IDE. The title bar includes 'Output', 'Serial Monitor', and a close button. Below the title bar is a message input field with the placeholder text 'Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem101')'. To the right of the input field are two dropdown menus: 'New Line' and '115200 baud'. The main area of the window displays a list of log messages in a monospaced font, each starting with a timestamp and followed by a status message and a time offset.

```
16:35:18.154 -> Heard unknown (203) @2992ms
16:35:20.560 -> Heard yes (207) @5424ms
16:35:21.652 -> Heard unknown (206) @6496ms
16:35:27.116 -> Heard unknown (203) @11968ms
16:35:30.085 -> Heard unknown (211) @14928ms
16:35:40.327 -> Heard yes (216) @7216ms
16:35:41.412 -> Heard unknown (203) @8304ms
```

Figure 1: Results on Serial Monitor